SST 411: TIME SERIES ANALYSIS

Comprehensive Lecture Notes — Text Edition

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*This edition was produced from the digital source of the notes. The lecture text was taken from the **text layer** of the document and re-set in this PDF; no optical recognition of any page was performed.*

*One thing must be said plainly: the source typesets nearly all of its mathematics — and several of its tables — as **images**. Images carry no text, so those parts are not present in the text layer and cannot be extracted with the words around them. Every such point is marked in this edition with a shaded note giving the page of the original on which the formula or table appears, e.g.*

> *[equations set as images in the original — see page 9 of the source document]*

The page numbers in these notes refer to the 45 pages of the original document, so any formula can be looked up there directly. Everything that the text layer does contain — all definitions, explanations, remarks, worked-example statements, data tables and assessment questions — is reproduced here in full and in the original order.

Original title: "SST 411: Comprehensive Lecture Notes on Time Series Analysis". Course: SST 411 — Time Series Analysis, 3 credit hours, pre-requisite SST 305.

Course Information

Course code and title: SST 411: TIME SERIES ANALYSIS — 3 Credit Hours
Pre-requisite: SST 305

Purpose of the course

This course aims at equipping the learner with basic knowledge of theories needed in making predictions of future values using past values.

Expected learning outcomes

By the end of this course students will be able to: measure and/or study various components of the time series namely: trend, seasonal, cyclic and random components.

Course content

Concept of time series. Components of time series: trend, seasonal, cyclic and random. Methods of analyzing the components. Use of filters in time series analysis. Time series processes: purely random, random walk, moving average, autoregressive processes. Correlogram, periodogram and spectral analysis.

Time series analysis flow chart

Week	Topic
Week 1	Definition and types of time series. Objectives of time series analysis
Week 2	Components, goals and models of time series analysis
Week 3	Measurement of trend component

Week	Topic
Week 4 and 5	Filtering method
Week 6	Weighted moving average process
Week 7	Analyzing series that contains seasonal and cyclic variation
Week 8	Analyzing series that contains cyclic variation
Week 9 and 10	Probability models for time series
Week 11 & 12	Spectral analysis
Week 13	Examination

Mode of delivery

Virtual classes, class lectures, discussions, problem solving, exercises.

Assessment

Continuous Assessment: 30%. Final Exam: 70%.

Core reading materials

Chatfield C. (2016) *The Analysis of Time Series: An Introduction*. CRC Press, New York. ISBN 9780203491683.

Recommended reading materials

1. Gupta S. C. and Kapoor V. K. (1994). *Fundamentals of Applied Statistics*. Sultan Chand and Sons. ISBN 8170141516.
2. Henrick Madsen (2007). *Time Series Analysis*. CRC Press, NY. ISBN 9781420059670.
3. Douglas et al. (2015). *Introduction to Time Series Analysis and Forecasting*. John Wiley and Sons, New Jersey.
4. Wayne A. Fuller (1995). *Introduction to Statistical Time Series*. John Wiley and Sons. ISBN 0471552399.

Journals

The Journal of Statistical Methods and Applications, Springer.

LESSON ONE

1.1 Introduction

In this lesson we introduce the time series analysis by looking at the four types. We will also explain some definitions useful in the analysis of time series.

1.2 Learning outcomes

By the end of this lesson the learner will be able to:

1. Define time series.

2. List down and explain the four types of time series.
3. Distinguish between (i) discrete and continuous time series, and (ii) deterministic and stochastic time series.
4. State the objectives of time series analysis.

1.3 Definition of a time series

A time series is a collection of observations made sequentially with time. It is an arrangement of statistical data in accordance with occurrence of time. Examples occur in many fields ranging from economics to engineering.

A time series involves two variables, one of which is time. For example population, X_t, in different years, t.

1.4 Types of time series

There are four types of time series:

1. Economic time series. Many types of time series arise in economics. Examples include share prices on successive days, export totals in successive months, average income in successive months etc.

2. Physical time series. Many types of time series occur in the physical sciences particularly meteorology, marine science and geophysics. Examples are rainfall on successive days and air temperatures measured in successive hours, days or months.

3. Marketing time series. This is especially in commerce. For example, analysis of sales figures in successive weeks or months is an important problem in commerce. It is often important to forecast future sales so as to plan production.

4. Demographic time series. This is the type of time series that occur in the study of population. An example is the population of Kenya taken after every 10 years. Demographers want to predict changes in the population for as long as ten or twenty years into the future.

1.5 Some terminologies

Continuous and discrete time series

A time series is said to be continuous when the observations are made continuously with time, and discrete when observations are taken only at specific times, usually equally spaced. The latter is the type that will be considered in this course.

Deterministic and stochastic time series

A deterministic time series is a series that can be predicted exactly. This is the case where future values may be predicted fully from past observations. However, most time series are stochastic in that the future is partly determined by past values, so that exact predictions are impossible.

In this course we will consider discrete and stochastic time series.

1.6 Objectives of time series analysis

The objectives of analysing a time series include Description, Explanation, Prediction and Control.

Description

When presented with a time series, the first step in the analysis is usually to plot the data and obtain simple descriptive measures of the main properties of the series. Features such as trend and seasonal effects can sometimes be easily seen from such a plot. The graph will also enable one to look for unusual observations or outliers which do not appear to be consistent with the rest of the data.

Explanation

When observations are taken on two or more variables, it may be possible to use the variation in one time series to explain the variation in another time series. This may lead to a deeper understanding of the process that generates the given time series.

Prediction

Given an observed time series, one may want to forecast the future values of the series. This is a very important task in sales forecasting and in the analysis of economic and industrial time series.

Control

When a time series is generated which measures the quality of a manufacturing process, the aim of the analysis may be to control the process. A stochastic model is fitted to the series, future values of the series are predicted and the input process variables are adjusted so as to keep the process on target.

LESSON TWO

2.1 Introduction

This lesson will look at the four components, objectives and goals of time series analysis.

2.2 Learning outcomes

By the end of the lesson the learner will be able to:

1. Explain the four components of time series.
2. State the goals of time series analysis.
3. Distinguish between additive and multiplicative models.

2.3 The four components of time series

There are four components of time series:

i. Secular trend or simply **trend**

ii. **Seasonal** variations or effects

iii. **Cyclic** variations/effects

iv. **Random** or irregular variations/effects

Trend

Trend is the general tendency of the data to increase or decrease over a long period of time. For example an upward tendency would be seen in data pertaining to population, agricultural production, currency in circulation etc., while a downward tendency will be noticed in data of deaths from births, epidemics etc. as a result of advancement in medical sciences, better medical facilities, literacy and higher standards of living.

Apart from upward and downward tendencies, we also have certain series whose values fluctuate round a constant reading which does not change with time. For example, temperature readings of a particular place operate about a constant room temperature.

Seasonal variations

These are periodic and regular movements in a time series with period of less than one year. For example sales and profits in a store, prices and consumption of certain commodities all exhibit seasonal swings.

Seasonal variations may be attributed to two causes:

a. Those resulting from natural forces, like production of commodities that depend on seasons, and prices of agricultural products which go down during harvesting and go up during planting.

b. Those resulting from man-made forces, for example sales and profits of some departmental stores increase during festivities.

Examples of series related to seasonal rhythms are: issue of books from the library, daily traffic density, hourly number of telephone messages in a certain switchboard, marriages etc.

Cyclic variations

These are oscillatory movements in a time series with period of oscillation more than one year. For example, review of production and review of salaries.

Random or irregular variations

These are fluctuations that are not accounted for by trend, seasonal or cyclic variations. They are purely random, unforeseen, unpredictable and beyond human control. For example wars, earthquakes, floods, epidemics, famine etc.

2.4 Goals of the analysis of time series

The main problem in a time series analysis is to

i. identify the component at work (trend, seasonal, cyclic or random effects) and

ii. isolate, study, analyse and measure them (components) independently.

2.5 Models for time series

There are two types of models for time series: additive and multiplicative models.

Additive model. According to the additive model, a time series can be expressed as

$$X_t = T_t + S_t + C_t + e_t$$

where

- X_t = time series value at time t
- T_t = trend value at time t
- S_t = seasonal value at time t
- C_t = cyclic value at time t
- e_t = random value at time t

This model implies that seasonal forces (in different years), cyclic forces (in different cycles) and random forces (in long term periods) operate with equal absolute effects irrespective of the long term of the trend value.

Multiplicative model. According to the multiplicative model,

$$X_t = T_t \cdot S_t \cdot C_t \cdot e_t$$

This is so if we assume that various components in a time series operate proportionally to the general level of the series.

Remark

1. S_t, C_t, and e_t, instead of assuming positive and negative values, are indices fluctuating above or below unity, and the geometric means of S_t in a year, C_t in a cycle and e_t in a long period are unity.
2. The multiplicative model can be converted to an additive model of logarithmic values of the original time series, i.e.

$$\ln X_t = \ln T_t + \ln S_t + \ln C_t + \ln e_t$$

The time plot should be examined to see which model is likely to give the better description.

LESSON THREE

3.1 Introduction

This lesson looks at various methods of measuring/studying trend such as the graphical method, the method of curve fitting and the filtering method.

3.2 Learning outcomes

By the end of this lesson the learner will be able to:

1. Use the graphic method to study trend.
2. Use the method of curve fitting in measuring trend.

3.3 Graphic method

Plot X_t against t and obtain a free-hand smooth curve. This will help us to form an idea about the general trend of the series. Though a simple and flexible method to study trend, the method does not measure trend and it is subjective.

3.4 Method of curve fitting

There are various types of curves that can be used to describe trend in a given data.

a) Polynomials

These are fitted using the method of least squares. For example:

i) Fitting a straight line trend $X_t = a + bt$

The normal equations are

$$\sum_{t=1}^{n} X_t = na + b\sum_{t=1}^{n} t$$

$$\sum_{t=1}^{n} tX_t = a\sum_{t=1}^{n} t + b\sum_{t=1}^{n} t^2$$

which are solved simultaneously for a and b.

ii) Fitting a second degree polynomial $X_t = a + bt + ct^2$

The normal equations are:

$$\sum X_t = na + b\sum t + c\sum t^2$$

$$\sum tX_t = a\sum t + b\sum t^2 + c\sum t^3$$

$$\sum t^2X_t = a\sum t^2 + b\sum t^3 + c\sum t^4$$

(In the source these three blocks of normal equations are set as equations in the image part of the page; see original p. 9.)

b) Exponential curve

$$X_t = ab^t$$

Putting logs we have

$$\log X_t = \log a + t\log b$$

which is linear in logs, i.e.

$$Y_t = A + Bt \qquad \text{where } Y_t = \log X_t, \quad A = \log a, \quad B = \log b.$$

The method of least squares is used to estimate A and B and consequently a and b.

c) Growth curves

Growth curves cannot be fitted by the method of least squares since the number of parameters is greater than the number of variables. We will use one of the methods, i.e. the method of three selected points, to fit growth curves.

i) Modified exponential,

$$X_t = a + bc^t \qquad \cdots(i)$$

We take 3 ordinates X_1, X_2 and X_3 (say) corresponding to three equidistant values of time t_1, t_2 and t_3 (say) such that $t_2 - t_1 = t_3 - t_2$.

Substituting these values in (i) we get

$$X_1 = a + bc^{t_1}, \qquad X_2 = a + bc^{t_2}, \qquad X_3 = a + bc^{t_3} \qquad \cdots(ii)$$

Subtracting the equations we get

$$X_2 - X_1 = c^{t_1}(c^{t_2 - t_1} - 1)b, \qquad X_3 - X_2 = c^{t_2}(c^{t_3 - t_2} - 1)b \qquad \cdots(iii)$$

$$\Rightarrow \quad c^{t_2 - t_1} = \frac{X_3 - X_2}{X_2 - X_1} = \frac{X_3 - X_2}{X_2 - X_1}$$

Substitute c in (iii) to get b and then substitute b and c in (ii) to get a.

ii) Gompertz curve,

$$X_t = ab^{c^t}$$

Taking natural logs on both sides we get

$$\ln X_t = \ln a + c^t \ln b$$

or

$$Y_t = A + Bc^t$$

with $Y_t = \ln X_t$, $A = \ln a$ and $B = \ln b$.

Hence the curve reduces to a modified exponential in logs.

iii) Logistic curve,

$$X_t = \frac{k}{1 + e^{a + bt}}, \qquad \text{where } a, b \text{ and } k \text{ are constants}$$

Consider 3 ordinates X_1, X_2 and X_3 (say) corresponding to three equidistant values of time t_1, t_2 and t_3 (say) such that $t_2 - t_1 = t_3 - t_2$. Then

$$X_1 = \frac{k}{1 + e^{a + bt_1}}, \qquad X_2 = \frac{k}{1 + e^{a + bt_2}}, \qquad X_3 = \frac{k}{1 + e^{a + bt_3}} \qquad \cdots(i)$$

$$a + bt_1 = \ln\left(\frac{k}{X_1} - 1\right) = \log\left(\frac{k - X_1}{X_1}\right)$$

$$a + bt_2 = \log\left(\frac{k - X_2}{X_2}\right), \qquad a + bt_3 = \log\left(\frac{k - X_3}{X_3}\right) \qquad \cdots(ii)$$

Subtracting these equations we get

$$b(t_3 - t_2) = \log\left[\frac{(k - X_3)X_2}{(k - X_2)X_3}\right] \qquad \text{and} \qquad b(t_2 - t_1) = \log\left[\frac{(k - X_2)X_1}{(k - X_1)X_2}\right]$$

Equating, since $t_3 - t_2 = t_2 - t_1$, we get

$$\frac{(k - X_3)X_2}{(k - X_2)X_3} = \frac{(k - X_2)X_1}{(k - X_1)X_2}$$

Solve k and substitute in (ii) to get b, and then substitute in (i) to get a.

3.5 Example 1

Below are given figures of production (in thousand tonnes) of a sugar factory:

Year	2003	2005	2006	2007	2008	2009	2012
Production (thousand tonnes)	67	88	94	85	91	98	90

i. Fit a straight line by the method of least squares and tabulate the trend values.

ii. Eliminate trend. What components of the time series are left over?

iii. What is the monthly increase/decrease in the production of sugar?

iv. Estimate the production of sugar in 2010.

3.6 Example 2

Given 3 selected points $X_1 = 58.8$, $X_2 = 138.6$, $X_3 = 251.8$ and corresponding to equidistant time intervals $t_1 = 2$, $t_2 = 30$ and $t_3 = 58$:

Fit (i) modified exponential and (ii) logistic curves by the method of three selected points.

3.7 Assessment

1. Fit a quadratic trend curve for the following data. Use $t = \text{year} - 2003$.

Year	2000	2001	2002	2003	2004	2005	2006
Number of production units, X_t	125	128	133	135	134	136	132

2. Use the method of three selected points $X_1 = 55$, $X_2 = 138$ and $X_3 = 251$ corresponding to t_1, t_2 and t_3 to fit a logistic curve $X_t = \frac{k}{1 + e^{a + bt}}$ where a, b and k are constants. Fit also the modified exponential and the Gompertz curves.

> *[In the source, the statement of this question carries the logistic curve as an image; see original p. 13.]*

LESSON FOUR

4.1 Introduction

Filtering is another method of measuring trend which will be looked into in this lesson.

4.2 Learning outcomes

By the end of this lesson the learner will be able to:

- 4.2.1 Define: (i) stationary time series, (ii) auto-correlation functions.
- 4.2.2 Describe the filtering method: (i) the moving average filter, (ii) the difference filter as a method of measuring trend.

4.3 Stationary time series

Before looking at filtering as a method of measuring trend, we look at the concept of a stationary time series.

Most of the probability theory of the time series analysis is concerned with stationary time series.

Broadly speaking, a time series is said to be stationary if there is no systematic change in mean level (no trend), if there is no systematic change in variance, and if strictly periodic variations have been removed.

Remark

From this definition, for a time series to be stationary, it should not have trend, seasonal and cyclic variations.

Strict (or strong) sense stationarity definition

A time series is said to be strictly stationary if the joint distribution of the time series is the same as the joint distribution of $\{X_{t_1+h}, X_{t_2+h}, \ldots, X_{t_n+h}\}$ for all $t_1, t_2, \ldots, t_n$ and h, where h is the distance between observations.

In this case the probability structure of the time series should not change with time for it to be stationary in the strict sense.

In practice, it is often useful to define stationarity in a less restricted way than the above.

Weak (or 2nd order) sense stationarity definition

A time series process is said to be weakly stationary if its mean, $E(X_t)$, is a constant and its auto-covariance function does not depend on time t but depends only on the distance between observations.

In general, a time series is neither stationary in the strict sense nor in the weak sense. For purposes of analysis, we use transformations to convert non-stationary series to stationary series.

In what follows, the word stationary will mean stationary in the weak sense.

4.4 Example 3

Let $\{X_t\}$ be a time series given by

$$X_t = a + bt + e_t$$

where $\{e_t\}$ is a sequence of independent random variables.

Show that X_t is non-stationary and that $Y_t = \Delta X_t = X_t - X_{t-1}$, where Δ is the difference operator, is stationary.

> *[In the source the statement of this example, with the defining formula, is partly set as an image; see original pp. 14–15.]*

4.5 Autocorrelation function

To compare the basic properties of a time series, it is often useful to have a function that is not influenced by units of measurement. Such a function is the theoretical autocorrelation function (ACF), defined as

$$\rho(h) = \frac{C(h)}{C(0)}, \qquad \text{where } C(h) = \operatorname{cov}(X_t, X_{t+h}) = E\big[(X_t - \mu)(X_{t+h} - \mu)\big]$$

and $C(h)$, $\rho(h)$ are the covariance and correlation functions of successive observations at lag h (h distance apart). Both are even functions, i.e.

$$C(h) = C(-h), \qquad \rho(h) = \rho(-h).$$

Periodic functions and periodic time series

Periodic functions play an important role in the analysis of empirical time series. Periodic functions are functions of sine and cosine functions of the form

$$X_t = A\cos(\lambda t + \phi)$$

where $-\pi < \lambda < \pi$, the absolute value of A is the amplitude, λ is the frequency and ϕ is the phase angle. The frequency is the number of times that the function repeats itself in a period (interval) of length $2\pi/\lambda$. The phase angle is the shift parameter that determines the points where the function is zero.

4.6 Example

Define

$$X_t = A\cos(\lambda t) + B\sin(\lambda t)$$

where A and B are independent random variables.

Show that X_t is second order stationary.

> *[The equations of §4.5 and §4.6 are image objects in the source; see original pp. 15–16.]*

4.7 Filtering as a method of measuring trend

For the analysis of any time series, we have first to convert a non-stationary time series into a stationary series. This is done by use of filters. From mathematical analysis, any function X_t, under some general conditions, can be represented by a Fourier series, i.e. a series of sums of sine and cosine functions.

Let

$$X_t = e^{i\lambda t} = \cos\lambda t + i\sin\lambda t, \qquad -\pi < \lambda < \pi, \qquad i = \sqrt{-1}$$

be the original time series and

$$Y_t = \sum_{j=-\infty}^{\infty} a_j X_{t-j}$$

be the transformed series after applying a filter a_j on X_t. Then

$$Y_t = \sum_{j=-\infty}^{\infty} a_j e^{i\lambda(t-j)} = e^{i\lambda t}\sum_j a_j e^{-i\lambda j} = e^{i\lambda t}\sum_j a_j\left(\cos\lambda j - i\sin\lambda j\right)$$

$$= e^{i\lambda t}\left[A_1(\lambda) + iA_2(\lambda)\right] = A(\lambda)e^{i\lambda t}$$

$$A(\lambda) = A_1(\lambda) + iA_2(\lambda), \qquad A_1(\lambda) = \sum_j a_j\cos\lambda j, \qquad A_2(\lambda) = \sum_j a_j\sin\lambda j$$

$$g(\lambda) = \left|A(\lambda)\right|, \qquad \phi(\lambda) = \tan^{-1}\frac{A_2(\lambda)}{A_1(\lambda)}$$

$$A(\lambda) = g(\lambda)e^{i\phi(\lambda)}, \qquad Y_t = g(\lambda)e^{i(\lambda t + \phi(\lambda))}$$

Conclusion

Comparing Y_t with X_t we note that

i. the amplitude changes from 1 to $g(\lambda)$;

ii. the wavelength remains invariant (does not change);

iii. the phase angle changes from zero to $\phi(\lambda)$.

The function $A(\lambda)$ is called the transform function of the filter. $g(\lambda)$ is the gain function of the filter and $\phi(\lambda)$ is the phase function of the function.

4.8 The moving average filter

If in

$$Y_t = \sum_{j=-m}^{m} a_j X_{t-j}, \qquad a_j = \frac{1}{2m+1}$$

then this filter is called a $(2m+1)$-point moving average filter. That is

$$Y_t = \frac{1}{2m+1} \sum_{j=-m}^{m} X_{t-j}$$

If $m = 1$, then $2m + 1 = 3$. This is a 3-point moving average filter.

Where

$$A(\lambda) = \sum_{j=-m}^{m} a_j e^{-i\lambda j} = \frac{1}{2m+1} \sum_{j=-m}^{m} e^{-i\lambda j} = \frac{1}{2m+1} \cdot \frac{\sin\left(\frac{2m+1}{2}\lambda\right)}{\sin\left(\frac{\lambda}{2}\right)},$$

a pure real number.

Let

$$X_t = \cos\lambda t.$$

Hence

$$Y_t = \frac{1}{2m+1} \sum_{j=-m}^{m} \cos\lambda(t-j)$$

Therefore

$$Y_t = \frac{1}{2m+1} \cdot \frac{\sin\left(\frac{2m+1}{2}\lambda\right)}{\sin\left(\frac{\lambda}{2}\right)} \cos\lambda t$$

We note that this filter has only one effect on the original series: it changes the amplitude of the series from 1 to

$$g(\lambda) = \frac{1}{2m+1} \cdot \frac{\sin\left(\frac{2m+1}{2}\lambda\right)}{\sin\left(\frac{\lambda}{2}\right)}.$$

What is the effect of this filter on different components of X_t?

Case 1

In this case we consider the trend components of X_t and Y_t when

$$\lambda \to 0.$$

$$Y_t \to \frac{1}{2m+1} \sum_{j=-m}^{m} X_{t-j} \to X_t$$

which is the trend value of X_t. This is so because as m becomes large $\frac{\sin\left(\frac{2m+1}{2}\lambda\right)}{\sin\left(\frac{\lambda}{2}\right)}$ becomes small. Hence X_t is dominated by the trend as $\lambda \to 0$. For the trend value of Y_t, we have $Y_t = X_t$.

Therefore, the $(2m+1)$-point moving average filter has no effect on the trend.

Case 2

$Y_t = 0$, if we take $\lambda = \frac{2\pi k}{2m+1}$ where k is a constant of proportionality.

Therefore if we choose m such that $2m+1$ is proportional to the wavelength or period of oscillation, then the $(2m+1)$-point moving average filter will destroy the oscillation.

Effect of the moving average filter on e_t. If m is sufficiently large, then the error terms will average to zero.

Remark

1. *How do we choose the appropriate filter?* The answer to this question requires considerable knowledge of the frequency aspects of a time series. As noted above, a moving average filter will eliminate periodic movements if the extent (period) of the moving average is proportional to the wavelength or period of oscillation to be eliminated. Thus for monthly data, a 12-point moving average filter will remove seasonality. For quarterly data, the 4-point moving average filter will remove seasonality.
2. A moving average of period (extent) n is a series of successive averages of n terms starting with the 1st, 2nd, 3rd, etc. Thus the 1st moving average is the average of the first n terms, the 2nd moving average is the average of the n terms starting with the second term to the $(n+1)^{\text{th}}$ term, and so on.

> *[The working of §4.7 and §4.8 is set as equations in the image part of the source; see original pp. 16–19.]*

4.9 Example 4

The following data show daily sales of company X for 4 successive weeks.

Day	Mon	Tue	Wed	Thur	Fri	Sat	Sun
Week I	153	189	221	215	302	223	201
Week II	133	171	241	228	283	255	238
Week III	145	200	187	201	292	220	233
Week IV	111	170	243	178	248	202	163

Find the trend values using an appropriate moving average filter.

Solution. We note that the data consists of an oscillation of period of length 7. To eliminate this oscillation, we use a 7-point moving average filter. The trend values will start from Thursday week I to Thursday week IV. The trend value for Thursday week IV is obtained as

$$Y = \frac{1}{7}\left(X_{\text{Mon III}} + \cdots + X_{\text{Sun IV}}\right)$$

Other trend values are similarly obtained.

> *[The arithmetic of the trend values is set as equations in the image part of the source; see original p. 20.]*

4.10 Difference filter

Let X_t be the original time series. Define

$$Y_t = \Delta X_t = X_t - X_{t-1}$$

where Δ is the difference operator. Then

$$Y_t = X_t - X_{t-1}.$$

Now

$$X_t = e^{i\lambda t} \quad \Rightarrow \quad Y_t = e^{i\lambda t} - e^{i\lambda(t-1)} = e^{i\lambda t}\left(1 - e^{-i\lambda}\right) = A(\lambda)e^{i\lambda t}$$

where

$$A(\lambda) = 1 - e^{-i\lambda} = 1 - \cos\lambda + i\sin\lambda.$$

Hence

$$g(\lambda) = \left|A(\lambda)\right| = 2\sin\left(\frac{\lambda}{2}\right), \qquad \phi(\lambda) = \tan^{-1}\left[\frac{\sin\lambda}{1-\cos\lambda}\right] = \frac{\lambda}{2} - \frac{\pi}{2}.$$

Conclusions

i. The amplitude changes from 1 to $2\sin\left(\frac{\lambda}{2}\right)$.

ii. The frequency remains the same.

iii. The phase angle changes from 0 to $\frac{\lambda}{2} - \frac{\pi}{2}$.

Case 1: Trend in the original series was $X_t = a + bt$; trend in $Y_t = 0$. Therefore this filter removes trend from X_t.

Case 2: The amplitude varies between 0 and 2. The wavelength remains invariant and the phase angle changes.

Effect on e_t. Suppose e_t is such that $E(e_t) = 0$, $V(e_t) = \sigma^2$ and $\text{cov}(e_t, e_{t+h}) = 0$. Then the new series is

$$Y_t = e_t - e_{t-1}$$

with

$$E(Y_t) = 0, \qquad V(Y_t) = 2\sigma^2, \qquad \text{cov}(Y_t, Y_{t-1}) = -\sigma^2.$$

The correlation between Y_t and Y_{t-1} is $-\frac{1}{2}$. Therefore differencing introduces correlation between the error terms.

> *[The derivation of §4.10 is set as equations in the image part of the source; see original pp. 20–22.]*

4.11 Assessment

1. Consider the process $X_t = \mu + e_t$ where e_t is a purely random process. Obtain its mean and auto-covariance function and comment on its stationarity.

> *[The defining equation is an image in the source; see original p. 22.]*

2. The following table shows profit (thousand Ksh) for four years given on a quarterly basis.

Year	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
2001	75	60	54	59
2002	86	65	63	78
2003	91	73	67	85
2004	104	81	75	96

Find the trend values using an appropriate moving average model.

3. The following table shows the population (in millions) of a certain country in Europe.

Census Year	1949	1959	1969	1979	1989	1999	2009
Population (X_t)	25.0	25.1	27.9	31.9	36.1	43.9	54.7

Fit an exponential trend to the above data, where $t = \text{year} - 1949$, by the method of least squares and estimate the population for 2019.

LESSON FIVE

5.1 Introduction

This lesson introduces the weighted moving average process in determining trend.

5.2 Learning outcome

By the end of this lesson the learner will be able to determine trend values using the method of weighted moving average process.

5.3 Determination of trend by weighted moving average process

Suppose we have decided on the degree of the polynomial to fit data to determine trend. Instead of fitting the polynomial to the entire data, we could fit it to a smaller convenient set of points and keep moving to determine the trend. For example, if the trend curve is cubic, then instead of fitting this curve to the entire data, we could conveniently fit it to a set of four or five points and keep on moving to determine the trend. This method of determining trend is called the weighted moving average method. The procedure of determining trend is equivalent to taking linear combinations of the observations with coefficients which can be worked out.

5.4 Example 5

A cubic polynomial in t is fitted to 5 consecutive points of a time series. Estimate the trend value corresponding to the middle point of the set as a weighted moving average of the observed data. Hence calculate the trend values of the following data:

t	1	2	3	4	5	6	7	8	9	10	11	12
X_t	19.38	19.26	18.43	17.28	16.84	15.42	16.17	18.74	18.27	16.92	17	18.19

Solution. Let the 5 consecutive points be $X_{-2}, X_{-1}, X_0, X_1, X_2$ and the cubic polynomial be

$$X_t = a_0 + a_1 t + a_2 t^2 + a_3 t^3.$$

The moving average of the 5 consecutive points is X_0, which is equal to a_0 when $t = 0$ in the polynomial. To find a_0 we use the method of least squares. The normal equations are:

$$\sum_{t=-2}^{2} X_t = 5a_0 + a_1 \sum t + a_2 \sum t^2 + a_3 \sum t^3 \qquad \cdots(i)$$

$$\sum_{t=-2}^{2} tX_t = a_0 \sum t + a_1 \sum t^2 + a_2 \sum t^3 + a_3 \sum t^4 \qquad \cdots(ii)$$

$$\sum_{t=-2}^{2} t^2 X_t = a_0 \sum t^2 + a_1 \sum t^3 + a_2 \sum t^4 + a_3 \sum t^5 \qquad \cdots(iii)$$

$$\sum_{t=-2}^{2} t^3 X_t = a_0 \sum t^3 + a_1 \sum t^4 + a_2 \sum t^5 + a_3 \sum t^6 \qquad \cdots(iv)$$

From (i) and (iii) we have

$$\sum X_t = 5a_0 + 10a_2, \qquad \sum t^2 X_t = 10a_0 + 34a_2.$$

Solving simultaneously,

$$a_0 = \frac{1}{35}\left(12\sum_{-2}^{2} t^2 X_t - 17\sum_{-2}^{2} X_t\right)$$

$$\therefore \quad a_0 = \frac{1}{35}\left(-3X_{-2} + 12X_{-1} + 17X_0 + 12X_1 - 3X_2\right)$$

which is the trend value at $t = 0$.

This formula finds trend values by moving average of extent 5 when a polynomial of degree 3 is fitted. The numbers

$$-\frac{3}{35}, \quad \frac{12}{35}, \quad \frac{17}{35}, \quad \frac{12}{35}, \quad -\frac{3}{35}$$

are called **weights**. They are symmetric about the centre and sum up to unity.

The first trend value is at $t = 3$ (middle of the first 5 values). Hence

$$a_0 = \frac{1}{35}\left(-3(19.38) + 12(19.26) + 17(18.43) + 12(17.28) - 3(16.84)\right)$$

For $t = 4$,

$$a_0 = \frac{1}{35}\left(-3(19.26) + 12(18.43) + 17(17.28) + 12(16.84) - 3(15.42)\right)$$

Other values are obtained in a similar manner.

> *[The normal equations (i)–(iv) and the weight formula are image equations in the source; see original pp. 24–26.]*

5.5 Assessment — Example 6

A quadratic polynomial in t is fitted to 7 consecutive points of a time series. Estimate the trend value corresponding to the middle point of the set as a weighted moving average of the observed data. Hence calculate the trend values of the following data:

t	1	2	3	4	5	6	7	8	9	10	11	12
X_t	19.38	19.26	18.43	17.28	16.84	15.42	16.17	18.74	18.27	16.92	17	18.19

a. A quadratic polynomial in t is fitted to 9 consecutive points of a time series. Estimate the trend value corresponding to the middle point of the set as a weighted average of the observed values. Hence find the trend values of the following time series data:

Year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
Sales	10	20	40	30	60	50	40	20	60	25

LESSON SIX

6.1 Introduction

Seasonal variations are exhibited by most of the business and economic variables, and their study is associated with the following reasons:

i. to isolate the seasonal variations in order to determine the effects of seasons on the size of the variable, and

ii. to eliminate seasonality so as to determine what would be the value of the variable if there were no seasonal swings.

The determination of seasonal effects is of paramount importance in planning business efficiency or planning a production programme.

Three common models in use are:

Model	Form
Additive	$X_t = T_t + S_t + e_t$
Multiplicative	$X_t = T_t S_t e_t$
Mixed	$X_t = T_t S_t + e_t$

For the study of seasonal variation, the data must be given for the parts of the year, namely monthly or quarterly, weekly, daily or hourly. There are different methods for measuring seasonal variations. We discuss the method of simple averages and the ratio to trend method.

6.2 Learning outcomes

By the end of the lesson the learner should be able to compute seasonal effects using:

1. The method of simple averages.

2. The ratio to trend method.

6.3 Method of simple averages

Step 1: Arrange the data by parts of the year, i.e. months or quarters.

Step 2: Compute the averages of the parts of the year, e.g.

$\bar{X}_i$, $(i = 1, 2, \ldots, 12$ for monthly data; $i = 1, 2, 3, 4$ for quarterly data, etc.)

Step 3: Compute the overall average $\bar{\bar{X}}$, the average of the averages of the parts of the year.

Step 4: Obtain the seasonal indices

$$S_i = \frac{\bar{X}_i}{\bar{\bar{X}}} \times 100 \quad \text{(multiplicative case)} \quad \text{or} \quad S_i = \frac{\bar{X}_i - \bar{\bar{X}}}{\bar{\bar{X}}} \times 100 \quad \text{(additive case)}$$

Note:

$$\sum_{i=1}^{4} S_i = 400 \quad \text{for quarterly data, and} \quad \sum_{i=1}^{12} S_i = 1200 \quad \text{for monthly data.}$$

The normalized/adjusted indices are respectively

$$\text{Adj}\, S_i = \frac{S_i}{\sum_{i=1}^{4} S_i} \times 400 \quad \text{and} \quad \text{Adj}\, S_i = \frac{S_i}{\sum_{i=1}^{12} S_i} \times 1200$$

> *[The formulas of Steps 2–4 and the adjustment formulas are image equations in the source; see original pp. 27–28.]*

6.4 Example 7

The data below give the average quarterly prices of a commodity for four years.

Year	1st quarter	2nd quarter	3rd quarter	4th quarter
2007	40.3	44.8	46.0	48.0
2008	50.1	53.1	55.3	59.5
2009	47.2	50.1	52.1	55.2
2010	55.4	59.0	61.6	65.3

Calculate the seasonal indices assuming a multiplicative model.

Solution.

	1st quarter	2nd quarter	3rd quarter	4th quarter
$\bar{X}_i$	48.25	51.75	53.75	57
$S_i = \frac{\bar{X}_i}{\bar{\bar{X}}} \times 100$	91.57	98.22	102.01	108.18

where

$$\bar{\bar{X}} = \frac{1}{4} \sum_{i=1}^{4} \bar{X}_i = 52.69.$$

$$\sum_{i=1}^{4} S_i = 399.98 \approx 400,$$

hence no need of normalizing.

The seasonal index for each quarter indicates how that quarter performs relative to the year as a whole. For example, for the 1st quarter, $S_i = 91.57\%$ means that sales for the 1st quarter are 91.57% of the average for the full year, or sales are 8.43% below the yearly quarterly average.

Remark

This method of simple averages is based on the assumption that the data do not contain any trend and cyclic components, and consists in eliminating irregular components by averaging the quarterly or monthly values over years. These assumptions are not in general true, since most economic time series have trend. For series having substantial trend and cyclic effects, methods such as the ratio to trend method, ratio to moving average and link relative methods are used.

6.5 Ratio to trend method

This method improves the method of simple averages and is based on the assumption that seasonal variation for any given month or quarter is a constant factor of the trend.

Steps:

1. Obtain the trend values by the method of least squares.
2. Eliminate trend by expressing the original values as percentages of the trend values. Assuming a multiplicative model, these percentages will contain the seasonal, cyclic and irregular components.
3. Eliminate the cyclic and irregular components by averaging the percentages for different months or quarters to get indices of seasonal variations.
4. Adjust the above indices to get a total of 1200 for monthly data or 400 for quarterly data by multiplying them throughout by a normalization factor given by

$$k = \frac{1200}{\sum_{i=1}^{12} S_i} \quad \text{for monthly data,} \qquad k = \frac{400}{\sum_{i=1}^{4} S_i} \quad \text{for quarterly data.}$$

Normalization removes the irregular components, leaving only the seasonal factor.

6.6 Example 8

Calculate the seasonal variations for the following data by the ratio to trend method.

Year	1st quarter	2nd quarter	3rd quarter	4th quarter
2000	30	40	36	34
2001	34	52	50	44
2002	40	58	54	48
2013	54	76	68	62

Year	1st quarter	2nd quarter	3rd quarter	4th quarter
2014	80	92	86	82

First we determine the trend values for the yearly average X_t by the method of least squares. We fit the trend line equation $X_t = a + bt$, taking the origin year as 2002, i.e. $t = \text{Year} - 2002$.

Year	t	Yearly average X_t	t^2	tX_t	Trend values
2000	−2	35	4	−70	32
2001	−1	45	1	−45	44
2002	0	50	0	0	56
2013	1	65	1	65	68
2014	2	85	4	170	80
Total	0	280	10	120	

The normal equations are

$$\sum X_t = na + b\sum t, \qquad \sum tX_t = a\sum t + b\sum t^2$$

Giving $a = 56$ and $b = 12$. Hence $X_t = 56 + 12t$.

The yearly increment is $b = 12$, so the quarterly increment is $12/4 = 3$, giving the following quarterly trend values:

Year	1st quarter	2nd quarter	3rd quarter	4th quarter
2000	27.5	30.5	33.5	36.5
2001	39.5	42.5	45.5	48.5
2002	51.5	54.5	57.5	60.5
2013	63.5	66.5	69.5	72.5
2014	75.5	78.5	81.5	84.5

Trend-eliminated values are then obtained by expressing the original values as percentages of the computed trend values, giving the following table:

Year	1st quarter	2nd quarter	3rd quarter	4th quarter
2000	109.1	131.1	107.5	93.1
2001	86.1	122.4	109.9	90.7
2002	77.7	106.4	93.9	70.3
2013	85.0	114.3	97.8	85.5
2014	106	117.1	105.5	97.0
Total	463.9	591.3	514.6	445.6
Average, S_i	92.78	118.26	102.92	89.12
Adj S_i	92.1	117.4	102.1	88.4

For the 1st quarter of 2000, the trend-eliminated value $= (30/27.5) \times 100 = 109.1$

.

> *[The normal-equation working and the trend arithmetic are image equations in the source; see original pp. 29–31.]*

6.7 Assessment — Example 9

Use the data on Example 7 to find the seasonal indices using the ratio to trend method.

LESSON SEVEN

7.1 Introduction

This lesson discusses one method of determining cyclic variations.

7.2 Learning outcomes

By the end of this lesson the learner will be able to determine cyclic variations given a time series data.

7.3 Measurement of cyclic variations

One method of measuring cyclic fluctuations of a time series consists of eliminating trend and seasonal variation and then removing the random component by averaging. Using the multiplicative model, the following steps are involved in the calculation of cyclic variations:

1. Calculate the trend values T_t and seasonal indices S_t.
2. Divide X_t by $T_t \times S_t$.
3. The resulting value gives C_t, the cyclic component.

7.4 Example 10

Consider the information in Example 7. Obtain the indices of cyclic variations from the data provided in Example 7.

Consider the table of trend-eliminated values in Example 7. Seasonal effects are eliminated by dividing the trend-eliminated values by the adjusted seasonal indices and multiplying by 100, giving y.

$$\text{Seasonal eliminated value} = \left(\frac{\text{trend eliminated value}}{\text{Adj } S_i}\right) \times 100 = y, \qquad \text{Cyclic variation} = y - 100$$

Year	1st quarter	2nd quarter	3rd quarter	4th quarter
2000	118.5	111.2	104.8	107.0
2001	93.1	103.5	107.1	104.3
2002	84.0	90.2	91.5	80.8
2013	93	96.9	95.3	98.3
2014	114.6	99.3	102.8	111.5

For the 1st quarter of 2000, the seasonal eliminated value $= (109.1/92.1) \times 100 = 118.5$.

Cyclic variation $= y - 100$:

Year	1st quarter	2nd quarter	3rd quarter	4th quarter
2000	18.5	11.2	4.8	7.0
2001	−6.9	3.5	7.1	4.3
2002	−16	−9.8	−8.5	−19.2
2013	−7	−3.1	−4.7	−1.7
2014	14.6	−0.7	2.8	11.5

Note that variations change after a period of more than one year.

Remark

Harmonic analysis provides a sophisticated method for determining the cyclic component of the time series.

LESSON EIGHT

8.1 Introduction

Probability models are used to analyse time series containing random variations. We will look at four special time series models, looking at their stationarity/conditions for which they are stationary, and also other properties.

8.2 Learning outcomes

By the end of the lesson the learner should be able to find the auto-covariance and auto-correlation functions of:

1. a purely random process,
2. a random walk process,
3. a moving average process.

8.3 Purely random process

The process $\{X_t: t = 0, \pm 1, \pm 2, \ldots\}$ is called a **purely random process** or **white noise** if it consists of a sequence of uncorrelated random variables $\{e_t\}$ with mean μ and variance σ^2.

This is the simplest of all time series models and corresponds to the case where the process has no memory, in the sense that the value of the process at time t is uncorrelated with all past values up to the time $t - 1$. For this process, $X_t = e_t$ with

$$E(X_t) = \mu \qquad \text{and} \qquad V(X_t) = \sigma^2.$$

The auto-covariance function is given by

$$\gamma(h) = \text{cov}(X_t, X_{t+h}) = E\big[(X_t - \mu)(X_{t+h} - \mu)\big] = E\big[X_t X_{t+h}\big] - E(X_t)E(X_{t+h})$$

$$= \begin{cases} \sigma^2, & h = 0 \\ 0, & h \neq 0 \end{cases}$$

$$\gamma(h) = \text{cov}(e_t, e_{t+h}) = E[e_t e_{t+h}] - E(e_t)E(e_{t+h}) = E(e_t e_{t+h}) - \mu^2$$

which is a function of h alone. Hence X_t is 2nd order stationary.

The auto-correlation function is

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \begin{cases} 1, & h = 0 \\ 0, & h \neq 0 \end{cases}$$

8.4 Random walk

Suppose $\{e_t\}$ is a discrete purely random process with mean μ and variance σ^2. Then a process $\{X_t\}$ is said to be a **random walk** if

$$X_t = X_{t-1} + e_t \qquad \text{and} \qquad X_0 = 0.$$

Hence

$$X_1 = e_1, \qquad X_2 = X_1 + e_2 = e_1 + e_2, \qquad X_3 = X_2 + e_3 = e_1 + e_2 + e_3, \qquad \ldots$$

$$X_t = e_1 + e_2 + \cdots + e_t = \sum_{j=1}^{t} e_j.$$

Now

$$E(X_t) = t\mu,$$

which is not a constant. Hence the random walk is **not stationary**.

It can be made stationary by the first difference, i.e.

$$\Delta X_t = Y_t = X_t - X_{t-1} = \sum_{j=1}^{t} e_j - \sum_{j=1}^{t-1} e_j = e_t.$$

$$E(Y_t) = \mu, \qquad \text{a constant,}$$

and

$$\gamma(h) = \text{cov}(Y_t, Y_{t+h}) = \text{cov}(e_t, e_{t+h}) = E(e_t e_{t+h}) - E(e_t)E(e_{t+h}) = \text{cov}(e_t, e_{t+h}) + \sigma^2 - \mu^2 = \sigma^2$$

which does not depend on t. Hence Y_t is 2nd order stationary.

The best known examples of time series that behave like random walks are share prices on successive days.

> *[The auto-covariance working for §8.3 and §8.4 is set as equations in the image part of the source; see original pp. 34–35.]*

8.5 Moving average (MA) process

Let $\{e_t\}$ be a purely random process with mean 0 and variance σ^2. Then the process $\{X_t\}$ is said to be a **moving average of order** q, MA(q), if

$$X_t = \beta_0 e_t + \beta_1 e_{t-1} + \cdots + \beta_q e_{t-q} = \sum_{j=0}^{q} \beta_j e_{t-j}$$

where $\{\beta_j\}$ are constants.

Now

$$E(X_t) = 0 \qquad \text{and} \qquad V(X_t) = \sum_{j=0}^{q} \beta_j^2 V(e_{t-j}) = \sigma^2 \sum_{j=0}^{q} \beta_j^2.$$

Auto-covariance and auto-correlation functions of MA(q)

$$\gamma(h) = \text{cov}(X_t, X_{t+h}) = E[X_t X_{t+h}]$$

$$= E\left[\left(\sum_{j=0}^{q} \beta_j e_{t-j}\right)\left(\sum_{k=0}^{q} \beta_k e_{t+h-k}\right)\right] = \sum_{j=0}^{q}\sum_{k=0}^{q} \beta_j \beta_k E\left(e_{t-j} e_{t+h-k}\right)$$

$$= \begin{cases} \sigma^2 \sum_{j=0}^{q-h} \beta_j \beta_{j+h}, & h = 0, 1, 2, \ldots, q \\ 0, & h > q \end{cases}$$

The auto-correlation function is

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \begin{cases} \frac{\sum_{j=0}^{q-h} \beta_j \beta_{j+h}}{\sum_{j=0}^{q} \beta_j^2}, & h = 0, 1, 2, \ldots, q \\ 0, & h > q \end{cases}$$

$$= \begin{cases} 1, & h = 0 \\ \frac{\sum_{j=0}^{q-h} \beta_j \beta_{j+h}}{\sum_{j=0}^{q} \beta_j^2}, & h = 1, 2, \ldots, q \\ 0, & h > q \end{cases}$$

MA(q) is 2nd order stationary since $\gamma(h)$ does not depend on t but on h, and $E(X_t) = 0$, a constant.

Remark

a. **Definition.** A correlogram is a graph of $\rho(h)$ against lag h. It is useful in interpreting a set of auto-correlation coefficients by providing an objective criterion for exploring the nature of the internal structure of the time series.

b. The correlogram for MA(q) is a damped oscillation and cuts off at lag q.

c. A moving average process cannot be identified uniquely for a given $\rho(h)$.

d. MA processes are mainly used in econometrics.

8.6 Example 11

Consider two MA(1) processes given by

$$A\colon X_t = e_t + \theta e_{t-1} \qquad\qquad B\colon X_t = e_t + \frac{1}{\theta} e_{t-1}$$

For A,

$$\rho(h) = \begin{cases} 1, & h = 0 \\ \frac{\theta}{1+\theta^2}, & h = \pm 1 \\ 0, & h > 1 \end{cases}$$

For B,

$$\rho(h) = \begin{cases} 1, & h = 0 \\ \frac{\theta}{1+\theta^2}, & h = \pm 1 \\ 0, & h > 1 \end{cases}$$

Hence $\rho(h)$ is not unique.

> *[The auto-covariance derivations of §8.5 and Example 11 are image equations in the source; see original pp. 35–37.]*

LESSON NINE

9.1 Introduction

This is another special probability model for time series. We will also look at the auto-covariance and auto-correlation properties of the model.

Let $\{e_t\}$ be a purely random process with mean 0 and variance σ^2. Then the process $\{X_t\}$ is said to be an **autoregressive process of order** p, AR(p), if

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \cdots + \alpha_p X_{t-p} + e_t = \sum_{j=1}^{p} \alpha_j X_{t-j} + e_t, \qquad \alpha_p \neq 0.$$

9.2 Learning outcomes

By the end of this lesson the learner will be able to derive the auto-correlation function of

1. an AR(1) process,
2. an AR(2) process.

9.3 The AR(1) process

One of the simplest and most useful stationary time series models is the AR(1), defined by

$$X_t = \alpha X_{t-1} + e_t, \qquad |\alpha| < 1,$$

with $E(e_t) = 0$ and

$$\text{cov}(e_t, e_{t'}) = \begin{cases} \sigma^2, & t = t' \\ 0, & t \neq t' \end{cases}$$

By successive substitution an AR(1) can be converted to an MA process of infinite order as follows. Define a backward shift operator B such that $B^j X_t = X_{t-j}$. Then for AR(1),

$$X_t - \alpha X_{t-1} = e_t \quad \Rightarrow \quad X_t - \alpha B X_t = e_t \quad \Rightarrow \quad (1 - \alpha B) X_t = e_t$$

$$\Rightarrow \quad X_t = (1 - \alpha B)^{-1} e_t = \left(1 + \alpha B + \alpha^2 B^2 + \cdots\right) e_t$$

$$\Rightarrow \quad X_t = e_t + \alpha e_{t-1} + \alpha^2 e_{t-2} + \cdots = \sum_{j=0}^{\infty} \alpha^j e_{t-j},$$

which is an infinite MA process.

$$E(X_t) = 0 \qquad \text{and} \qquad V(X_t) = V(e_t)\left(1 + \alpha^2 + \alpha^4 + \cdots\right) = \frac{\sigma^2}{1-\alpha^2}.$$

Next,

$$\gamma(h) = \text{cov}(X_t, X_{t+h}) = E\left[X_t X_{t+h}\right] = \sum_j \sum_k \alpha^j \alpha^k E\left(e_{t-j} e_{t+h-k}\right) = \sigma^2 \sum_{j=0}^{\infty} \alpha^j \alpha^{j+h} = \frac{\sigma^2 \alpha^h}{1-\alpha^2}$$

which does not depend on t. Hence X_t is 2nd order stationary, and

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \alpha^h.$$

Another way of finding $\rho(h)$ of an AR(1) is to assume that the process is stationary and that $E(X_t) = 0$:

$$\begin{aligned} E\left[X_{t-h} X_t\right] &= \alpha E\left[X_{t-h} X_{t-1}\right] + E\left[X_{t-h} e_t\right] \\ \Rightarrow \quad \gamma(h) &= \alpha\gamma(h-1) + 0 \\ \Rightarrow \quad \gamma(h) &= \alpha\gamma(h-1) \\ = \alpha^2\gamma(h-2) &= \alpha^3\gamma(h-3) = \alpha^4\gamma(h-4) = \cdots \\ = \alpha^h\gamma(h-h) &= \alpha^h\gamma(0), \qquad \text{on successive substitution} \\ \Rightarrow \quad \rho(h) &= \frac{\gamma(h)}{\gamma(0)} = \alpha^h. \end{aligned}$$

Exercise. Plot the correlogram of AR(1) when $\alpha > 0$ and when $\alpha < 0$.

9.4 The AR(2) process

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + e_t, \qquad |\alpha_1|, |\alpha_2| < 1.$$

Assuming $E(X_t) = 0$, and multiplying throughout by X_{t-h} and taking expectation, we have

$$\begin{aligned} E\left[X_{t-h} X_t\right] &= \alpha_1 E\left[X_{t-h} X_{t-1}\right] + \alpha_2 E\left[X_{t-h} X_{t-2}\right] + E\left[X_{t-h} e_t\right] \\ \Rightarrow \quad \gamma(h) &= \alpha_1\gamma(h-1) + \alpha_2\gamma(h-2) \\ \Rightarrow \quad \rho(h) &= \alpha_1\rho(h-1) + \alpha_2\rho(h-2), \qquad h > 0 \qquad \cdots(1) \end{aligned}$$

The set of equations given by (1) is called the **Yule–Walker equations**.

Let $\rho(h) = m^h$ be a trial solution of (1):

$$\begin{aligned} m^h &= \alpha_1 m^{h-1} + \alpha_2 m^{h-2} \\ \Rightarrow \quad m^h\left(m^2 - \alpha_1 m - \alpha_2\right) &= 0 \end{aligned}$$

Since $m^h \neq 0$, the auxiliary equation is

$$m^2 - \alpha_1 m - \alpha_2 = 0 \qquad \cdots(2)$$

the general solution of (1) is

$$\rho(h) = A_1\pi_1^h + A_2\pi_2^h \qquad \cdots(3)$$

where π_1 and π_2 are the roots of (2), and A_1 and A_2 are constants to be determined.

The condition for this process to be stationary is that

$$|\pi_1|, |\pi_2| < 1, \qquad \text{i.e.} \qquad \left|\frac{\alpha_1 \pm \sqrt{\alpha_1^2 + 4\alpha_2}}{2}\right| < 1.$$

Case 1: $\alpha_1^2 + 4\alpha_2 \geq 0$

In this case π_1 and π_2 are real roots. We wish to find A_1 and A_2. From (3),

$$\rho(0) = A_1 + A_2 \quad \Rightarrow \quad A_1 + A_2 = 1 \qquad \cdots(i)$$

since $\rho(0) = \frac{\gamma(0)}{\gamma(0)} = 1$.

From (1),

$$\rho(1) = \alpha_1\rho(0) + \alpha_2\rho(1)$$
$$\Rightarrow \quad \rho(1) = \frac{\alpha_1}{1-\alpha_2} \qquad \cdots(ii)$$

From (3),

$$\rho(1) = A_1\pi_1 + A_2\pi_2.$$

Using (i) and (ii) we get the values of A_1 and A_2; when these are substituted in (3), the auto-correlation function of an AR(2) is obtained.

The correlogram of an AR(2) will decay exponentially if π_1 and π_2 are real roots.

Case 2: $\alpha_1^2 + 4\alpha_2 < 0$

In this case π_1 and π_2 are complex roots:

$$\pi_1 = \frac{\alpha_1 + i\sqrt{-\alpha_1^2 - 4\alpha_2}}{2} = re^{i\theta}$$

where

$$r = \sqrt{-\alpha_2} \qquad \text{and} \qquad \theta = \tan^{-1}\left(\frac{\sqrt{-\alpha_1^2 - 4\alpha_2}}{\alpha_1}\right) = \tan^{-1}\left(\frac{\sqrt{4\alpha_2 + \alpha_1^2}}{\alpha_1}\right).$$

Similarly,

$$\pi_2 = \frac{\alpha_1 - i\sqrt{-\alpha_1^2 - 4\alpha_2}}{2} = re^{-i\theta}.$$

The general solution of the Yule–Walker equation (3) now becomes

$$\rho(h) = A_1\left(re^{i\theta}\right)^h + A_2\left(re^{-i\theta}\right)^h = r^h\left(A_1e^{i\theta h} + A_2e^{-i\theta h}\right)$$
$$= r^h\left[A_1(\cos\theta h + i\sin\theta h) + A_2(\cos\theta h - i\sin\theta h)\right]$$
$$= r^h\left[(A_1 + A_2)\cos\theta h + i(A_1 - A_2)\sin\theta h\right]$$
$$= r^h\left[\cos\theta h + (A_1 - A_2)\, i\sin\theta h\right], \qquad \text{using } (i) \qquad \cdots(iii)$$

From (iii),

$$\rho(1) = r\left[\cos\theta + i(A_1 - A_2)\sin\theta\right] = \frac{\alpha_1}{1-\alpha_2}, \qquad \text{using } (ii)$$
$$\Rightarrow \quad (A_1 - A_2)\, i\sin\theta = \frac{\alpha_1}{1-\alpha_2}\cdot\frac{1}{r} - \cos\theta.$$

Substituting this value in (iii) you get the acf of an AR(2) process. This will be in the form

$$\rho(h) = Ar^h\sin(\theta h + \phi)$$

whose correlogram is a damped oscillation.

> *[The algebra of §9.3–§9.4 is set as equations in the image part of the source; see original pp. 38–42.]*

9.5 Example 12

Consider an AR(2) process given by

$$X_t = \frac{1}{3}X_{t-1} + \frac{2}{9}X_{t-2} + e_t, \qquad \text{where } e_t \sim N\left(0, \sigma^2\right).$$

Is this process stationary? If so, find its acf and sketch its correlogram for $h = 0, 1, 2, \ldots, 20$.

Solution.

$$E\left[X_{t-h}X_t\right] = \frac{1}{3}E\left[X_{t-h}X_{t-1}\right] + \frac{2}{9}E\left[X_{t-h}X_{t-2}\right] + E\left[X_{t-h}e_t\right]$$

$$\Rightarrow \quad \gamma(h) = \frac{1}{3}\gamma(h-1) + \frac{2}{9}\gamma(h-2)$$

$$\Rightarrow \quad \rho(h) = \frac{1}{3}\rho(h-1) + \frac{2}{9}\rho(h-2) \qquad \cdots(i)$$

Let $\rho(h) = m^h$:

$$m^h = \frac{1}{3}m^{h-1} + \frac{2}{9}m^{h-2} \quad \Rightarrow \quad m^h\left(m^2 - \frac{1}{3}m - \frac{2}{9}\right) = 0$$

The auxiliary equation is

$$m^2 - \frac{1}{3}m - \frac{2}{9} = 0 \quad \Rightarrow \quad \pi_1 = -\frac{1}{3}, \quad \pi_2 = \frac{2}{3}.$$

X_t is second order stationary since

$$|\pi_1| = \frac{1}{3} < 1, \qquad |\pi_2| = \frac{2}{3} < 1.$$

The general solution to the Yule–Walker equation is

$$\rho(h) = A_1\pi_1^h + A_2\pi_2^h = A_1\left(-\frac{1}{3}\right)^h + A_2\left(\frac{2}{3}\right)^h$$

$$\Rightarrow \quad \rho(0) = A_1 + A_2 = 1$$

and

$$\rho(1) = -\frac{1}{3}A_1 + \frac{2}{3}A_2.$$

From (i),

$$\rho(1) = \frac{1}{3}\rho(0) + \frac{2}{9}\rho(1) \quad \Rightarrow \quad \rho(1) = \frac{3}{7},$$

$$\Rightarrow \quad A_1 = \frac{5}{21}, \qquad A_2 = \frac{16}{21},$$

giving the acf as

$$\rho(h) = \frac{5}{21}\left(-\frac{1}{3}\right)^h + \frac{16}{21}\left(\frac{2}{3}\right)^h.$$

The correlogram decays exponentially.

9.6 Example 13

Consider an AR(2) process given by

$$X_t = X_{t-1} - \frac{1}{2}X_{t-2} + e_t, \qquad \text{where } e_t \sim N\left(0, \sigma^2\right).$$

Is this process stationary? If so, find its acf and sketch its correlogram for $h = 0, 1, 2, \ldots, 20$.

Solution.

$$\rho(h) = \rho(h-1) - \frac{1}{2}\rho(h-2)$$

The auxiliary equation is

$$m^2 - m + \frac{1}{2} = 0 \quad \Rightarrow \quad \pi_1 = \frac{1}{2} + \frac{1}{2}i, \quad \pi_2 = \frac{1}{2} - \frac{1}{2}i.$$

X_t is second order stationary since

$$|\pi_1| = |\pi_2| = \frac{1}{\sqrt{2}} < 1.$$

$$r = \frac{1}{\sqrt{2}}, \qquad \theta = \tan^{-1}(1) = 45^\circ = \frac{\pi}{4}$$

$$\Rightarrow \quad \pi_1 = re^{i\theta}, \qquad \pi_2 = re^{-i\theta}.$$

The general solution to the Yule–Walker equation is

$$\rho(h) = A_1\pi_1^h + A_2\pi_2^h = r^h\left(A_1e^{i\theta h} + A_2e^{-i\theta h}\right)$$

$$= \left(\frac{1}{2}\right)^{h/2}\left[(A_1 + A_2)\cos\frac{h\pi}{4} + i(A_1 - A_2)\sin\frac{h\pi}{4}\right] \qquad \cdots(*)$$

$$\rho(1) = \rho(0) - \frac{1}{2}\rho(1) \quad \Rightarrow \quad \rho(1) = \frac{2}{3}$$

From (*),

$$i(A_1 - A_2)\cdot\frac{1}{2}\cdot\frac{1}{\sqrt{2}} = \frac{2}{3}\cdot\frac{1}{\sqrt{2}}\cdot\frac{1}{1} - \cos\frac{\pi}{4} \quad \Rightarrow \quad i(A_1 - A_2) = \frac{1}{3}$$

$$\Rightarrow \quad \rho(h) = \left(\frac{1}{2}\right)^{h/2}\left[\cos\frac{h\pi}{4} + \frac{1}{3}\sin\frac{h\pi}{4}\right]$$

The correlogram is a damped oscillation.

> *[The working of Examples 12 and 13 is set as equations in the image part of the source; see original pp. 42–45.]*

9.7 Assessment

Consider the AR(2) processes given by

i. $X_t = \frac{1}{12}X_{t-1} + \frac{1}{12}X_{t-2} + e_t$

ii. $X_t = \frac{5}{16}X_{t-1} + \frac{5}{16}X_{t-2} + e_t$

where e_t is a purely random process with mean 0 and variance σ^2. Are these processes stationary? If so, find their autocorrelation functions.

> *[The two AR(2) models in this question appear as images in the source; see original p. 45.]*

Note on the End of the Notes

The flow chart lists “Week 11 & 12: Spectral analysis” as a further topic; the forty-five pages of these notes end with the assessment for Lesson Nine (autoregressive processes), so spectral analysis, the periodogram and the correlogram construction for spectral work are not covered in the material transcribed here.